

## KINSHASA N'DJILI INTL, Congo, the Democratic Republic of the

WMO: 642100

Lat: 4.386S

Lon: 15.445E

Elev: 1027

StdP: 14.16

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 7             | 66.0       | 67.7 | 59.7                          | 79.4 | 73.8 | 60.8 | 82.6 | 74.3 | 15.3                  | 77.3 | 13.9 | 77.6 | 2.9                      | 180  | 0.296 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 3             | 16.2                   | 93.5            | 77.2 | 92.2 | 76.9 | 91.3 | 76.6 | 79.1                | 88.9 | 78.6 | 88.1 | 78.1 | 87.2 | 7.8                  | 0    |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 76.8                            | 145.3 | 83.6 | 75.6 | 139.2 | 82.6 | 75.4 | 138.3 | 82.4 | 43.7          | 89.4 | 42.9 | 89.4 | 42.2 | 87.7 | 87.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 14.5              | 12.4 | 11.7 | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |
|                   |      |      | 59.1                       | 82.0 | 3.0                | 1.6 | 56.9                                               | 83.2 | 55.2       | 84.1 | 53.5       | 85.0 | 51.3       | 86.1 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 79.5   | 80.5  | 81.2  | 81.9  | 81.8  | 80.7  | 76.8  | 75.4  | 76.7  | 79.3  | 80.3  | 80.1  | 79.8 |
|                                                                               | DBStd       | 3.17   | 2.29  | 2.92  | 2.36  | 2.27  | 2.23  | 2.48  | 2.26  | 2.53  | 2.47  | 2.65  | 2.47  | 2.36 |
|                                                                               | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | CDD50       | 10779  | 944   | 873   | 990   | 954   | 950   | 804   | 788   | 829   | 879   | 940   | 903   | 925  |
|                                                                               | CDD65       | 5304   | 479   | 453   | 525   | 504   | 485   | 354   | 323   | 364   | 429   | 475   | 453   | 460  |
|                                                                               | CDH74       | 44893  | 4290  | 4321  | 5099  | 4662  | 4176  | 2310  | 1862  | 2633  | 3739  | 4235  | 3775  | 3791 |
| CDH80                                                                         | 15484       | 1461   | 1668  | 2083  | 1862  | 1449  | 553   | 403   | 761   | 1283  | 1493  | 1277  | 1191  |      |
| Wind                                                                          | WSAvg       | 5.3    | 4.5   | 5.0   | 5.1   | 4.9   | 4.7   | 5.1   | 5.7   | 6.5   | 6.4   | 5.6   | 5.0   | 4.8  |
| Precipitation                                                                 | PrecAvg     | 58.5   | 6.1   | 5.6   | 6.8   | 8.3   | 5.6   | 0.2   | 0.0   | 0.3   | 1.3   | 6.1   | 10.6  | 7.9  |
|                                                                               | PrecMax     | 75.7   | 13.5  | 12.0  | 12.0  | 18.4  | 15.5  | 1.9   | 0.3   | 2.0   | 2.9   | 11.4  | 18.1  | 14.5 |
|                                                                               | PrecMin     | 42.8   | 1.1   | 1.1   | 2.4   | 4.0   | 0.5   | 0.0   | 0.0   | 0.0   | 0.3   | 2.7   | 5.6   | 3.3  |
|                                                                               | PrecStd     | 9.0    | 2.8   | 2.7   | 2.4   | 3.3   | 2.7   | 0.4   | 0.1   | 0.6   | 0.7   | 2.4   | 3.0   | 3.0  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 92.8  | 94.8  | 95.2  | 95.0  | 93.4  | 89.8  | 87.9  | 89.9  | 93.0  | 93.6  | 91.8  | 91.7 |
|                                                                               |             | MCWB   | 77.5  | 77.4  | 77.3  | 78.1  | 77.7  | 74.3  | 73.2  | 72.6  | 74.3  | 76.0  | 76.2  | 76.5 |
|                                                                               | 2%          | DB     | 91.0  | 91.8  | 93.3  | 93.1  | 91.3  | 87.4  | 85.9  | 88.0  | 90.5  | 91.5  | 90.0  | 89.8 |
|                                                                               |             | MCWB   | 76.9  | 77.2  | 77.4  | 77.7  | 77.5  | 74.6  | 72.2  | 71.9  | 73.8  | 75.4  | 76.0  | 76.3 |
|                                                                               | 5%          | DB     | 89.3  | 90.8  | 91.5  | 91.3  | 89.4  | 85.6  | 84.2  | 86.2  | 89.0  | 89.6  | 88.8  | 88.0 |
|                                                                               |             | MCWB   | 76.5  | 76.9  | 77.0  | 77.5  | 76.8  | 74.0  | 71.7  | 71.4  | 73.4  | 75.1  | 75.8  | 76.1 |
|                                                                               | 10%         | DB     | 87.4  | 88.7  | 89.6  | 89.4  | 87.5  | 83.1  | 82.3  | 84.4  | 86.8  | 87.6  | 87.0  | 86.2 |
|                                                                               |             | MCWB   | 76.1  | 76.3  | 76.6  | 77.0  | 76.5  | 73.1  | 71.0  | 70.8  | 73.0  | 74.8  | 75.7  | 76.1 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 79.2  | 79.7  | 79.8  | 79.7  | 79.5  | 77.3  | 75.0  | 74.4  | 76.5  | 78.0  | 78.4  | 78.6 |
|                                                                               |             | MCDB   | 88.1  | 89.6  | 90.6  | 90.4  | 88.8  | 84.6  | 84.3  | 86.2  | 87.2  | 88.6  | 87.2  | 86.8 |
|                                                                               | 2%          | WB     | 78.3  | 78.6  | 78.8  | 78.9  | 78.5  | 75.8  | 73.5  | 73.2  | 75.3  | 76.8  | 77.5  | 77.8 |
|                                                                               |             | MCDB   | 86.7  | 87.8  | 89.2  | 89.4  | 87.5  | 83.8  | 83.1  | 85.4  | 87.3  | 87.1  | 86.3  | 85.8 |
|                                                                               | 5%          | WB     | 77.5  | 77.9  | 78.1  | 78.2  | 77.8  | 74.9  | 72.5  | 72.2  | 74.3  | 76.1  | 76.9  | 77.1 |
|                                                                               |             | MCDB   | 85.5  | 86.8  | 87.9  | 88.2  | 86.4  | 82.8  | 81.6  | 83.6  | 85.9  | 86.0  | 85.5  | 85.1 |
|                                                                               | 10%         | WB     | 76.9  | 77.2  | 77.3  | 77.4  | 77.1  | 73.8  | 71.6  | 71.3  | 73.5  | 75.4  | 76.2  | 76.5 |
|                                                                               |             | MCDB   | 84.4  | 85.7  | 86.5  | 86.9  | 85.4  | 81.0  | 80.0  | 82.1  | 84.5  | 84.9  | 84.3  | 83.9 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 14.1  | 15.2  | 16.2  | 16.5  | 14.6  | 13.6  | 13.9  | 15.3  | 14.9  | 14.7  | 14.3  | 13.6 |
|                                                                               |             | MCDBR  | 16.7  | 17.7  | 18.7  | 18.6  | 17.3  | 16.3  | 16.9  | 18.3  | 18.3  | 17.7  | 16.9  | 16.0 |
|                                                                               |             | MCWBR  | 6.0   | 6.2   | 6.6   | 6.7   | 6.7   | 6.6   | 7.0   | 7.0   | 6.5   | 5.9   | 5.8   | 5.9  |
|                                                                               | 5% WB       | MCDBR  | 14.8  | 16.2  | 17.2  | 17.5  | 15.8  | 14.5  | 15.1  | 16.7  | 16.4  | 16.1  | 15.7  | 14.5 |
|                                                                               |             | MCWBR  | 6.0   | 6.3   | 6.9   | 6.8   | 6.7   | 6.3   | 6.7   | 6.9   | 6.4   | 6.0   | 6.0   | 5.9  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.567  | 0.558 | 0.517 | 0.447 | 0.562 | 0.671 | 0.743 | 0.773 | 0.660 | 0.505 | 0.462 | 0.541 |      |
|                                                                               | taud        | 1.961  | 1.999 | 2.129 | 2.342 | 1.972 | 1.687 | 1.545 | 1.500 | 1.739 | 2.144 | 2.272 | 2.038 |      |
|                                                                               | Ebn at Noon | 250    | 253   | 260   | 272   | 232   | 201   | 189   | 191   | 222   | 264   | 276   | 255   |      |
|                                                                               | Edh at Noon | 62     | 60    | 52    | 41    | 56    | 73    | 85    | 93    | 75    | 51    | 45    | 57    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 1514   | 1512  | 1616  | 1570  | 1400  | 1264  | 1316  | 1348  | 1391  | 1362  | 1453  | 1411  |      |
|                                                                               | RadStd      | 65     | 70    | 83    | 82    | 66    | 64    | 58    | 100   | 57    | 63    | 58    | 67    |      |

## Historical Trends

|                            | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                            |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                            |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page